

Assignment 4 - CSE340

1. Design a 16 X 132 register file.

Note: You must label all the input and output pins

2. Why does the ALU control unit take the following bits as input?

Instruction[30, 14-12]

3. What is the significance of the Zero pin in the ALU?

4. Suppose you have a single instruction;

Add X21, X22, X23

Register read = 10s

Instruction Fetch = 50s

Memory Operation = 50s

Register write = 10s

ALU Operation = 20s

Suppose you have a pipelined setup and a single cycle setup,

Which one will run the instruction faster?

5. Modify the BEQ datapath so that it works for BNE.